INTRODUCTION

An earlier lesson established a derivative as a financial instrument that derives its performance from an underlying asset, index, or other financial variable, such as equity price volatility. Primary derivative types include a firm commitment in which a predetermined amount is agreed to be exchanged between counterparties at settlement and a contingent claim in which one of the counterparties determines whether and when the trade will settle. The following lessons define and compare the basic features of forward commitments and contingent claims and explain how to calculate their values at maturity.

Learning Module Overview

- Forwards, futures, and swaps represent firm commitments, or derivative contracts that require counterparties to exchange an underlying in the future based on an agreed-on price.
- Forwards are a flexible over-the-counter (OTC) derivative instrument, while futures are standardized and traded on an exchange with a daily settlement of contract gains and losses.
- Swap contracts are a firm commitment to exchange a series of cash flows in the future. Interest rate swaps are the most common type and involve the exchange of fixed interest payments for floating interest payments.
- Option contracts are contingent claims in which one of the counterparties determines whether and when a trade will settle. The option buyer pays a premium to the seller for the right to transact the underlying in the future at a pre-agreed exercise price.
- Option contract payoff and profit profiles are non-linear as the underlying price changes, as opposed to firm commitments, such as forwards, futures, and swaps, which are linear in underlying price changes.
- Market participants often create similar exposures to an underlying using firm commitments and contingent claims, although these derivative instrument types involve different payoff and profit profiles.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Which of the following statements correctly describes a difference between a forward contract and a futures contract?

   A. A forward contract sets an agreed-on price for buyer and seller, while a futures contract does not.
   B. A forward contract sets an agreed-on transaction date for the seller to deliver the underlying to the buyer, while a futures contract does not.
   C. A forward contract does not require daily settlement of gains and losses, while a futures contract does.

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C is correct. Futures contracts require daily settlement through the exchange clearinghouse mark-to-market process. Forward contracts are settled at their maturity date, although the two parties to the contract may customize alternative settlement procedures. A is incorrect because both forward and futures contracts set an agreed-on price for a future transaction. B is incorrect because both forwards and futures contracts include a maturity date when the underlying will be exchanged.

2. Identify which example fits each of the following firm commitments:

A. Futures contract purchaser	1. Agrees to make a single exchange in the future at a pre-agreed price under an OTC contract
B. Forward contract seller	2. Agrees to a single exchange in the future based on standardized terms set by an exchange
C. Fixed-rate payer on an interest rate swap	3. Agrees to a series of exchanges of interest fixed for floating interest payments

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Solution:

1. B is correct. A forward contract seller agrees to make a single exchange in the future at a pre-agreed price under an OTC contract.

2. A is correct. A futures contract purchaser agrees to a single exchange in the future based on standardized terms set by an exchange.

3. C is correct. A fixed-rate payer on an interest rate swap agrees to a series of exchanges of fixed for floating interest payments.

3. Identify which example fits each of the following contingent claims:

A. Put option purchaser	1. Seeks to gain from an increase in the underlying price
B. Call option purchaser	2. Allows the option to expire at maturity of the underlying price is above the exercise price
C. Both a put option purchaser and a call option purchaser	3. Pays an option premium to the option seller when the contract is agreed on

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Solution:

1. B is correct. A call option purchaser seeks to gain from an increase in the underlying price.

2. A is correct. A put option purchaser will allow an option to expire at maturity without exercise if the underlying price is above the exercise price.

3. C is correct. Both a put option purchaser and a call option purchaser will pay a premium to the option seller when the option contract is executed.

4. An option to buy an underlying security at an exercise price of USD45 in three months trades at a premium of USD6. After three months, the underlying trades at USD50. Which of the following responses correctly describes the profit/loss position of the option buyer and seller?

A. Option buyer earns USD5 profit, and option seller earns USD5 loss.
B. Option buyer earns USD1 loss, and option seller earns USD1 profit.
C. Option buyer earns USD5 profit, and option seller earns USD0.

Hide Solution

Solution:

this cash flow more than offsets the positive payoff. Thus, the option buyer's overall profit is a loss of USD1 (i.e., 5 – 6). For the option seller, the option position creates a negative payoff of –USD5, equal to –max(0, 50 – 45). However, the option seller received the option premium of USD6, so the overall profit is USD1 (i.e., 6 – 5). A is incorrect because the USD5 amount reflects the option payoff only, not profits and losses accounting for the option premium. C is incorrect because the buyer's profit incorrectly states the payoff only to the option position, not the profit. The seller's profit would be correct only if the underlying traded at 51, not 50.

5. A put option buyer earns a positive profit in which of the following conditions?

A. The price of the underlying at option expiration is less than the option's exercise price.
B. The price of the underlying at option expiration is greater than the option's exercise price.
C. The price of the underlying is less than the option's exercise price minus the option's premium.

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Solution:

C is correct. For a put option buyer to earn a positive profit, the underlying price must be sufficiently below the put option's exercise price such that (1) the put option can be exercised with a positive payoff and (2) the positive payoff is greater than the option premium paid. Thus, only if the underlying price falls below the exercise price minus the premium can this occur. A is incorrect because this condition only implies a positive payoff on the option but would include prices at which the payoff is not greater than the premium. B is incorrect because the put option would be out of the money and would generate zero payoff.

6. Which of the following positions on the same underlying benefit from opposite price movements in an underlying?

A. Long forward contract, short put option
B. Short forward contract, long put option
C. Short forward contract, short put option

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Solution:

C is correct. A short forward position benefits as the underlying price declines, while a short put benefits only when the underlying price increases. A is incorrect because both a long forward and a short put benefit from underlying price increases. B is incorrect because both a short forward and a long put option benefit from underlying price decreases.

Discuss Filter

Discussion

What do you want to discuss... (required)

PS in 4,shouldn't the option buyer just let the option expire since the net payoff incurs losses for him??

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